

Quiz-1 (Brief solution)

Notation: $Fea(P)$ is the feasible region of (P)

LI: Linearly Independent

No credit for answers without any justification or wrong justification

1. Consider the following problem (P):

$$\text{Min } -2x_1 + 3x_2$$

subject to

$$3x_1 - x_2 \leq b_1$$

$$4x_1 + x_2 \geq b_2$$

$$x_1 \geq 0, x_2 \geq 0.$$

- (a) If $b_1 = 6$ and $b_2 = 4$, give all the extreme directions of (P), and **check** whether (P) has an optimal solution by using the extreme directions. Give the optimal solution/s if there exists one.

Soln: The two extreme directions are $\mathbf{d} = [0, 1]^T$ and $\mathbf{d}' = [1, 3]^T$ and all positive scalar multiples of these two vectors.

Needs to be justified either from the picture or calculations.

Since $\mathbf{c}^T \mathbf{d} = 3 > 0$ and $\mathbf{c}^T \mathbf{d}' = 7 > 0$ so (P) has an optimal solution.

The problem has a unique optimal solution $[2, 0]^T$.

Needs to be justified.

[3.5]

If you got only 1 direction instead of 2 and argued accordingly then 1 mark deducted or this part.

- (b) Does (P) have an optimal solution for all $b_1, b_2 \in \mathbb{R}$? If yes justify, if not give an example to justify your claim.

Soln: (P) has optimal solution for all $b_1, b_2 \in \mathbb{R}$.

If $\mathbf{b} = [+, -]^T, [+, +]^T, [-, -]^T, [-, +]^T$ then any $x_1 = 0$ and $x_2 \geq \max \{|b_1|, |b_2|\}$, will give a feasible solution to (P). Since the set of directions does not depend on $\mathbf{b}$ and so remains unchanged, $\mathbf{c}^T \mathbf{d} = 3 > 0$, $\mathbf{c}^T \mathbf{d}' = 7 > 0$, so for any $\mathbf{b}$, (P) has optimal solution. [2.5]

No marks for this part if you have not shown the feasibility of (P) for all $b_1, b_2 \in \mathbb{R}$.

2. Consider the following problem (P):

$$\text{Min } \mathbf{c}^T \mathbf{x}$$

subject to $A_{2 \times 4} \mathbf{x} \leq \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$,

where $\mathbf{c}^T = [1, 2, 1, -5]$, $\mathbf{b} = [2, -1]^T$, the **first row** of A is $[-2, -4, 3, 2a]$, the **second row** of A is $[2, a, -6, a]$, and $0 \leq a \leq 1$.

- (a) If possible give two **extreme directions**, and a **direction** which is **not** an extreme direction, of $Fea(P)$.

Soln: One extreme direction is

- (i) $[1, 0, \frac{1}{3}, 0]^T$, which lies on the three LI hyperplanes $d_2 = 0, d_4 = 0, [2, a, -6, a]\mathbf{d} = 0$ defining D the set of directions of $Fea(P)$.

- (ii) $[1, 0, \frac{2}{3}, 0]^T$ is another extreme direction which lies on the three LI hyperplanes $d_2 = 0, d_4 = 0, [-2, -4, 3, 2a]\mathbf{d} = 0$ defining D the set of directions of $Fea(P)$.

All the calculations need to be shown.

For each $\mathbf{d}$, it needs to be shown that $A\mathbf{d} \leq 0$.

There are other alternate correct answers.

- (iii) Another direction which is not an extreme direction since it is the sum of the previous two extreme directions is $[2, 0, 1, 0]^T$. [3]

Also marks are deducted and you get only half the total credit, if your extreme directions are not fully specified for example if it depends on a which is unknown.

- (b) If possible give an $\mathbf{x}_0 \in Fea(P)$ and a $\mathbf{d} \in \mathbb{R}^4$, such that $\mathbf{x}_0$ lies on **exactly** three **LI** defining hyperplanes of $Fea(P)$, and $\mathbf{x}_0 + \alpha\mathbf{d}$ for some $\alpha \geq 0$, is an extreme point of $Fea(P)$ which also lies on the three **LI** hyperplanes on which $\mathbf{x}_0$ lies. Mention the extreme point and all the hyperplanes involved. Justify, if not possible.

Soln: The point which lies on exactly three LI hyperplanes defining $Fea(P)$ is

- (i) $\mathbf{x}_0 = [1, 0, \frac{4}{3}, 0]^T$ which lies on the hyperplanes $x_2 = 0, x_4 = 0, [-2, -4, 3, 2a]\mathbf{x} = 2$ and check that it belongs to $Fea(P)$. So if we pick a $\mathbf{d}$ orthogonal to the normals of the three LI hyperplanes on which $\mathbf{x}_0$ lies then we get the desired extreme point of $Fea(P)$. So for $\mathbf{d} = [-1, 0, -\frac{2}{3}, 0]^T$ we get $\mathbf{x}_0 + \alpha\mathbf{d} = [0, 0, \frac{2}{3}, 0]^T$ for $\alpha = 1$, which is an extreme point and it lies on the four LI hyperplanes given by $x_1 = 0, x_2 = 0, x_4 = 0, [-2, -4, 3, 2a]\mathbf{x} = 2$, three of which are common with $\mathbf{x}_0$. Check the feasibility of the point obtained.

- (ii) $\mathbf{x}_0 = [1, 0, \frac{1}{2}, 0]^T$ which lies on the hyperplanes $x_2 = 0, x_4 = 0, [2, a, -6, a]\mathbf{x} = -1$ and check that it belongs to $Fea(P)$. So if we pick a $\mathbf{d}$ orthogonal to the normals of the three LI hyperplanes on which $\mathbf{x}_0$ lies then we get the desired extreme point of $Fea(P)$. So for $\mathbf{d} = [-1, 0, -\frac{1}{3}, 0]^T$ we get $\mathbf{x}_0 + \alpha\mathbf{d} = [0, 0, \frac{1}{6}, 0]^T$ for $\alpha = 1$, which is an extreme point and it lies on the four LI hyperplanes given by $x_1 = 0, x_2 = 0, x_4 = 0, [2, a, -6, a]\mathbf{x} = -1$, three of which are common with $\mathbf{x}_0$. Check the feasibility of the point obtained.

There are other alternate correct answers.

Feasibility should be checked for each of the points obtained. All necessary calculations needs to be shown and NO partial marks for this part.

[4]

(c) Does (P) have an optimal solution?

Soln: Check that $\mathbf{d} = [1, 1, 1, 1]^T$ is a direction of $Fea(P)$ and $\mathbf{c}^T \mathbf{d} = -1 < 0$, since (P) is a minimization problem so (P) does not have an optimal solution. [3]

There are other alternate correct answers.

No partial marks for this part.